Tesla Financial Forecasting Project Report

Financial Data Analysis and Forecasting using Tesla Stock Data

Project Overview
This project focuses on financial data analysis and stock price forecasting using historical Tesla stock data. The objective was to analyze stock market trends, perform exploratory data analysis (EDA), create visualizations, and build forecasting models to predict future stock prices.

Data Cleaning
The dataset was collected using Yahoo Finance through the yfinance library. The preprocessing stage included:
- Handling missing values
- Converting Date column into datetime format
- Sorting the dataset chronologically
- Selecting relevant features
- Scaling stock prices using MinMaxScaler

Exploratory Data Analysis and Trend Analysis
EDA was performed to identify trends and volatility patterns in Tesla stock prices.
Key observations:
- Tesla stock showed high volatility.
- Long-term growth trends were observed.
- Trading volume fluctuated significantly during major market movements.
- Price-related features showed strong positive correlation.

Data Visualization
Several visualizations were created using Matplotlib and Power BI:
- Stock price trend line charts
- Trading volume charts
- Monthly average price charts
- Forecast comparison charts
- RMSE and MSE comparison bar charts

Forecasting Models
The following forecasting models were implemented:
1. SimpleRNN
2. LSTM
3. ARIMA (2,1,2)
4. Holt-Winters Exponential Smoothing

Performance Summary:
- SimpleRNN achieved the best short-term forecasting performance.
- LSTM showed stable long-term prediction capability.
- ARIMA captured linear trends moderately well.
- Holt-Winters struggled with Tesla’s nonlinear volatility.

Model Comparison Results
SimpleRNN (1-Day): RMSE = 19.61
LSTM (1-Day): RMSE = 19.87
LSTM (5-Day): RMSE = 34.93
LSTM (10-Day): RMSE = 44.40
SimpleRNN (5-Day): RMSE = 60.70

ARIMA (2,1,2): RMSE = 77.60
SimpleRNN (10-Day): RMSE = 82.70
Holt-Winters: RMSE = 98.79

Dashboard Development

An interactive Power BI dashboard was developed containing:
- Stock Market Overview Page
- Forecasting Dashboard
- Model Performance Dashboard

The dashboard included:
- KPI cards
- Actual vs predicted charts
- RMSE and MSE comparison visuals
- Interactive slicers and trend charts

Insights and Conclusions
- Deep learning models outperformed traditional statistical forecasting methods.
- SimpleRNN achieved the lowest forecasting error.
- LSTM provided stable long-term forecasting performance.
- Tesla stock prices were highly volatile and influenced by market dynamics.
- The project demonstrated the effectiveness of machine learning and dashboard analytics for financial forecasting.

Conclusion

This project successfully combined financial analytics, machine learning, deep learning, time-series forecasting, and dashboard visualization to create a complete end-to-end Tesla stock forecasting solution.